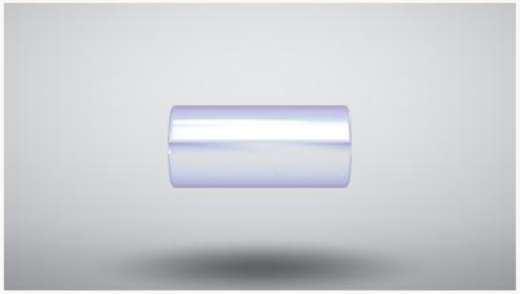
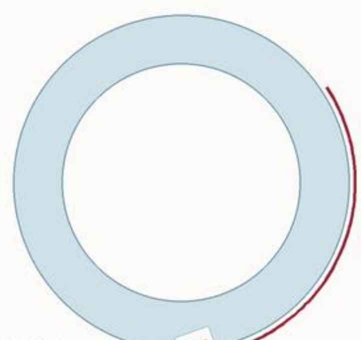
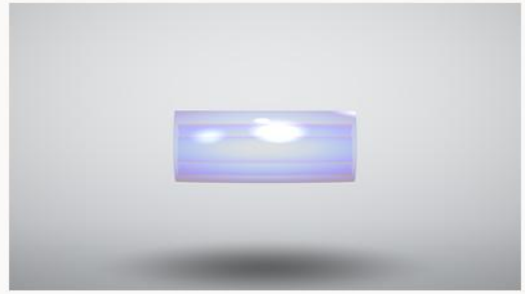
Glass tip — two ways

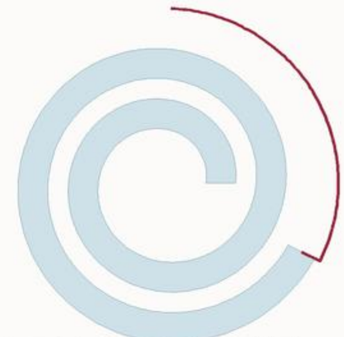
Same job, opposite answers. One is a tube with a slot in it. The other is a rolled sheet, and the roll is the slot. Sections at right, with the paper in.





the slot leans in, so the paper cannot pull straight back out





the free edge of the sheet leaves the slot, and it runs the whole length

Slotted tube

A slot down the length, cut in on a seventy-degree lean so it undercuts and holds the paper. Perforated screen across the bore.

Overall	**19 mm**
Outside	**ø 9 mm**
Bore	**ø 6.4, 1.3 wall**
Screen	**seven ø 1.25 holes**
Slot	**0.75 wide, 0.9 deep**
Mass	**≈ 1.4 g**

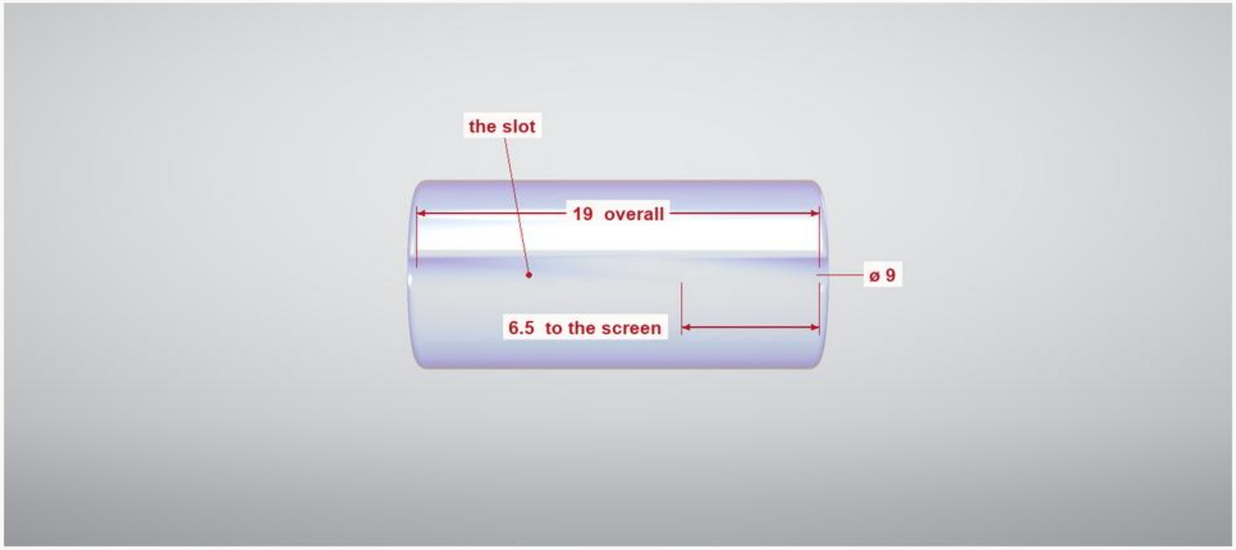
Rolled sheet

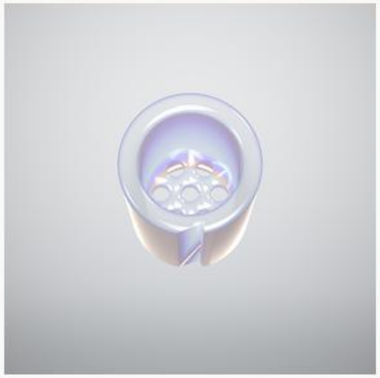
No tube at all - a thin sheet rolled into a 9 mm cylinder, and the stock cut into lengths. The gap between the wraps is the slot, it runs the whole piece, and the free edge is where the paper goes in.

Overall	**19 mm**
Outside	**ø 9 mm**
Sheet	**0.8 thick**
Gap	**0.55 between wraps**
Wraps	**about two and a half**
Mass	**≈ 1.15 g**

Glass tip

19 mm of clear 9 mm tube. Perforated screen across the bore, and a straight slot cut at a long oblique to start the roll.





Down the bore
seven holes, one centre

Overall length
19 mm

Outside diameter
9 mm

Bore / wall
6.4 / 1.3 mm

Screen
1.5 thick, 6.5 in

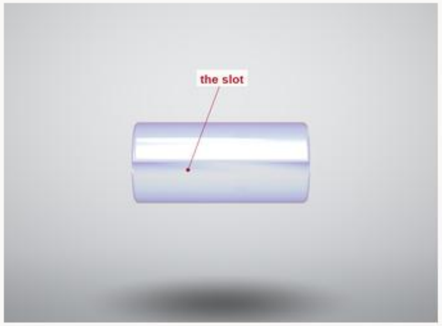
Screen holes
seven ø 1.25

Slot
0.75 wide, 0.9 deep

Slot rake
68° off the axis

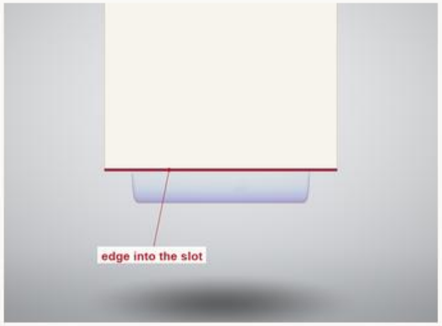
Glass
clear boro 3.3, ≈ 1.4 g

ROLLING



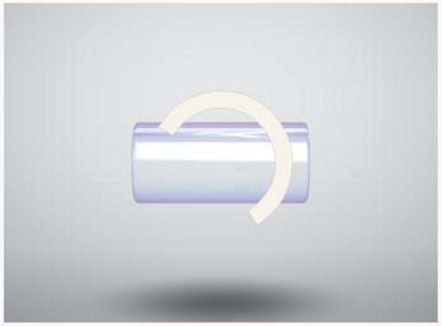
One

The slot runs the length of the tube, cut in on a lean.



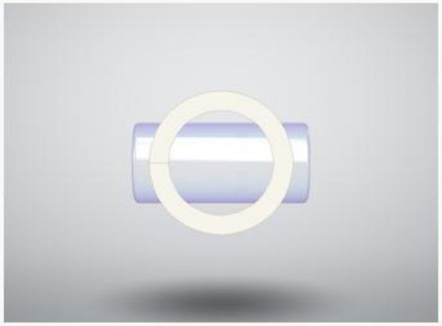
Two

Slip the leading edge of the paper into it. The slot holds the paper on its own.



Three

Roll the tip. The paper winds on, and the rake starts the wrap slightly off-square - the bias a cone wants.



Four

Wound on and filled. The screen keeps it from pulling through, and the tip washes and goes again.